pfi

Python for Fixed Income

COURSE 5C · RISK MODELS

Isolation Forest and Local Outlier Factor

WEEK 2 · DAY 2

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Today: unusual in several columns

Two datasets

2,000

curve days

12

planted errors

The Treasury curve is real and public domain; the book of 200 bonds is synthetic.

Three screens

Mahalanobis distance: far from the centre, in the data's own spread and correlation.
Isolation forest: cut off in few random splits.
Local outlier factor: sparser than its neighbours.

The labels only score. Every threshold and setting is chosen without them.

Names for a flag

An unusual curve day is listed as history, and explained by none of today's numbers. A flagged bond is a record to check.

Allowed names

a flag: a print to check

an unusual day, as history

a score, against planted labels

Never called

mispriced, rich, or cheap

an opportunity or a signal

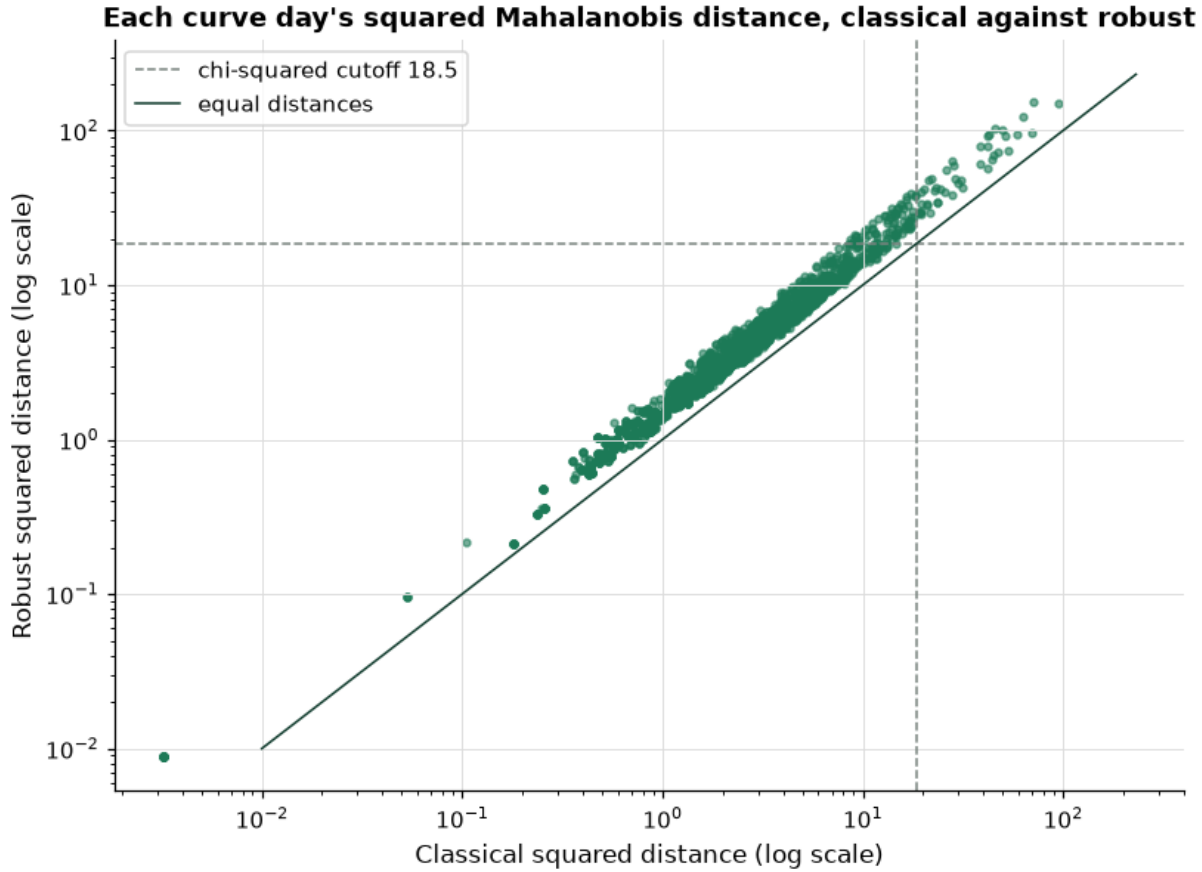
a correction by itself

No flag is a signal or a view on rates.

How the Mahalanobis distance works

The distance	(x - centre)' S^-1 (x - centre): each column in its own spread, correlations taken out
Why it matters	A day when tenors that move together moved apart is far, though each move was ordinary
The cutoff	For normal data, chi-squared with one degree of freedom per column
Classical	Centre and covariance from every day, the unusual ones included
Robust	From the most tightly packed subset of days, MinCovDet (mahalanobis_flags)

Classical against robust



44

days, classical

122

days, robust

Every day sits above the diagonal. The robust 10 Yr standard deviation is 4.012 bp against a classical 5.027 bp. The 78 days only the robust screen flags are the masked ones.

Cutoff 18.47; robust core 1,746 of 2,000 days. Fat tails: rank by it, never count by it.

Unusual days, as history

Day	2 Yr	5 Yr	10 Yr	30 Yr	Robust	Classical
2022-06-13	34.0	31.0	28.0	22.0	154.0	71.5
2020-03-06	-10.0	-9.0	-18.0	-31.0	150.6	95.3
2022-02-10	25.0	14.0	9.0	5.0	122.6	62.8
2022-06-15	-25.0	-23.0	-16.0	-6.0	103.3	45.7
2022-11-10	-27.0	-32.0	-30.0	-28.0	100.6	49.8
2020-03-17	11.0	17.0	29.0	29.0	97.5	70.0
2020-03-12	0.0	0.0	6.0	19.0	94.7	59.1
2020-02-28	-25.0	-22.0	-17.0	-14.0	94.2	42.6

Each tenor's change in bp, and the squared distances. Some days every tenor moved the same way; on others the short and long ends moved apart.

The table gives the changes. It does not say why the curve moved, and neither does the notebook.

Bonds whose fields disagree

CUSIP	Coupon	Price	Duration	Years	Error
99141LAA3	0.064	98.831	4.840	6.089	coupon typed as a decimal
99009JAC5	9.250	952.480	9.019	23.726	price times ten
99030EAE5	4.125	992.500	4.253	4.775	price times ten
99004EAA5	0.065	96.683	7.582	10.877	coupon typed as a decimal
99004EAB3	7.625	88.366	43.287	26.689	duration longer than the maturity
99144OAF3	5.750	104.801	15.543	10.995	duration longer than the maturity
99003DAG5	0.051	100.459	1.017	1.076	coupon typed as a decimal
99001BAA4	4.875	988.820	6.000	7.291	price times ten
99095RAA6	3.500	89.614	15.565	10.229	duration longer than the maturity
99093PAB0	0.064	101.051	3.170	3.669	coupon typed as a decimal
99129ZAB6	8.250	101.481	18.077	10.563	duration longer than the maturity
99062KAB8	6.250	1,030.160	1.416	1.533	price times ten

Four durations longer than the maturity, four coupons in percent typed as decimals, four prices times ten: planted in a copy by a seeded generator.

A price times ten is extreme in one column. A duration too long can be ordinary on its own: only the pair is wrong.

How isolation forest works

A tree	Pick a column at random, then a split point at random between its min and max; repeat
Isolation	Until every row stands alone
Path length	The splits it took to cut a row off; short means unusual
The forest	Path lengths averaged over many trees give each row's score
Settings	Trees, rows per tree, columns per tree, and contamination ('auto' or a share)

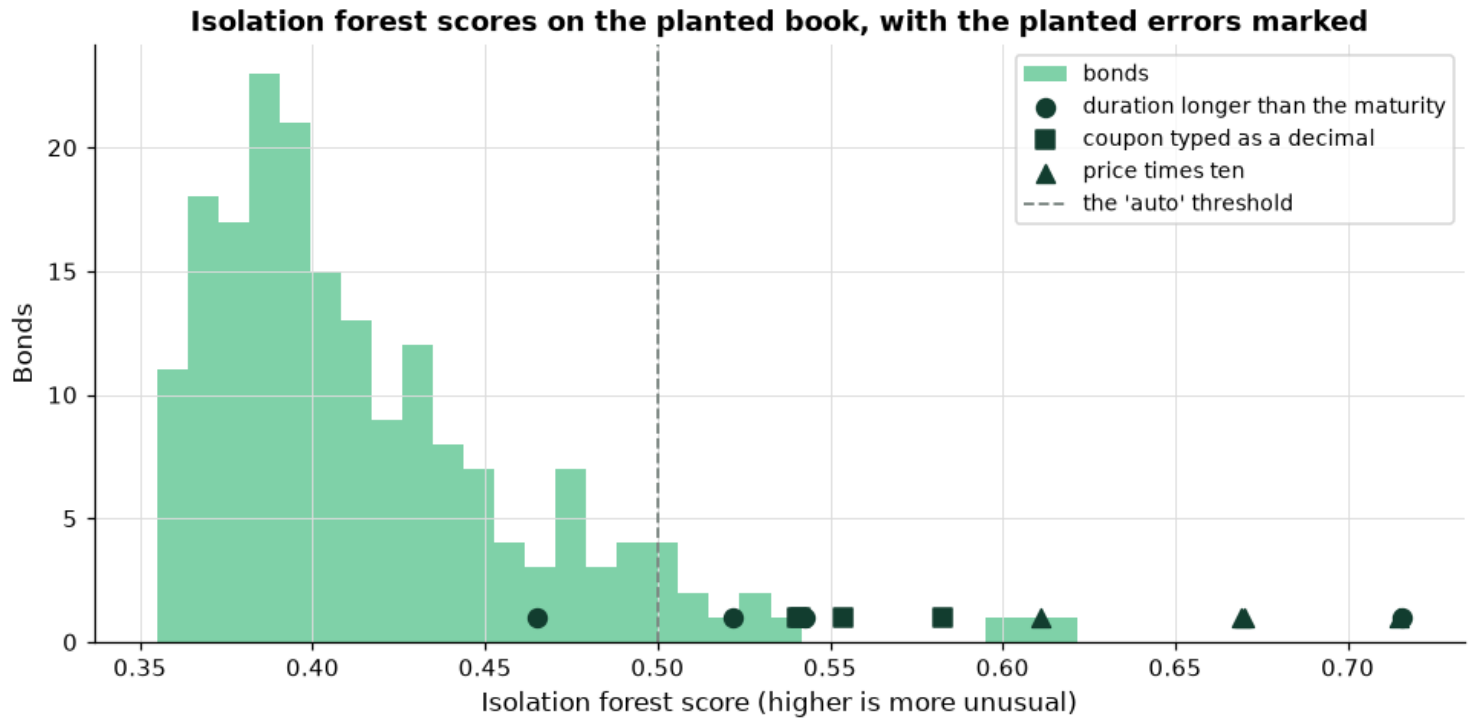
Every setting, swept

Setting	Value	Flags	Precision	Recall
n_estimators	25	30	37%	92%
	50	25	44%	92%
	100	22	50%	92%
	400	27	41%	92%
max_samples	32	61	18%	92%
	64	42	29%	100%
	128	29	38%	92%
	auto	22	50%	92%
max_features	0.4	26	31%	67%
	0.6	24	42%	83%
	1.0	22	50%	92%

More trees: the same recall, a few flags either way. Fewer rows per tree: more flags. Fewer columns per tree: lower recall.

Each setting trades what the screen finds against what it costs to check. The labels only score the sweep.

The forest's scores



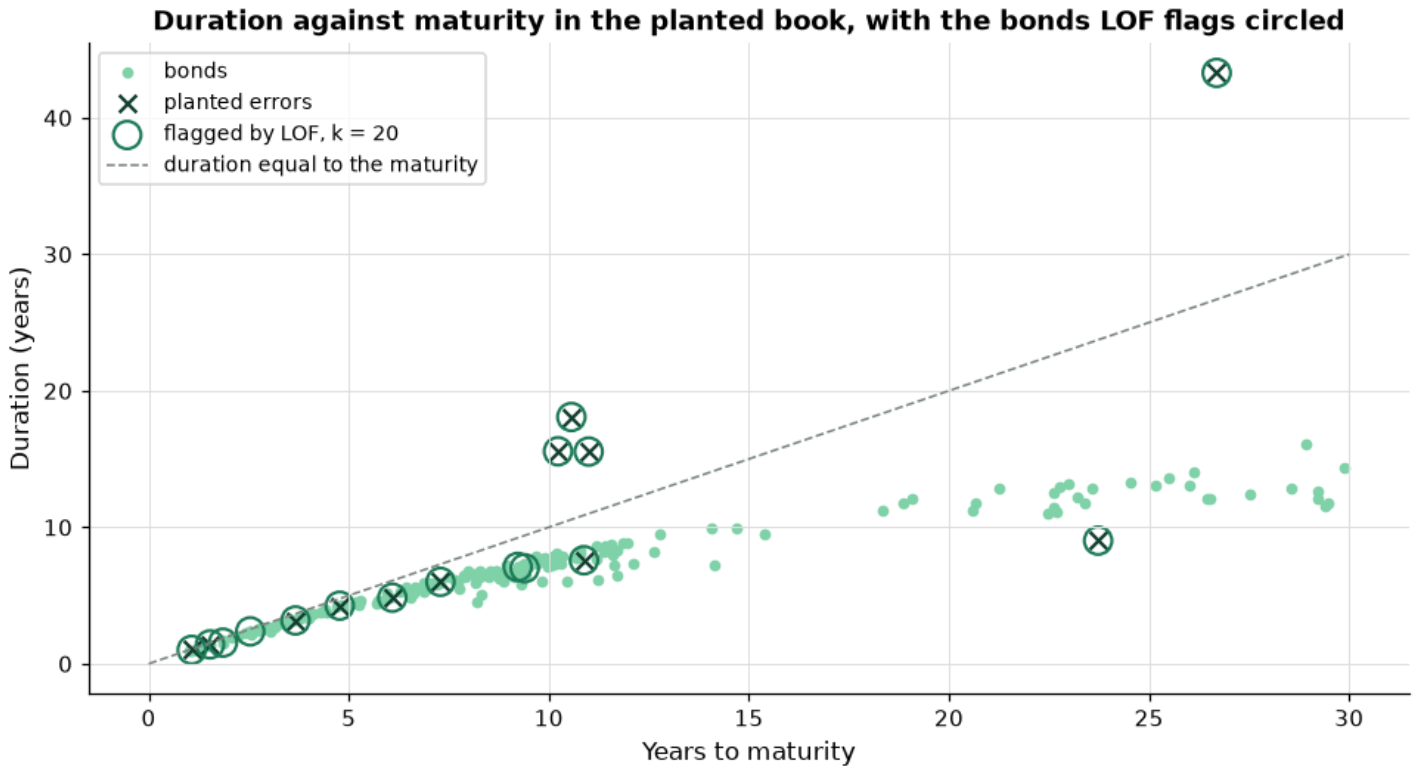
Prices times ten: far right. Decimal coupons: next. Durations longer than the maturity: spread out, one under the threshold.

A tree splits one column at a time: it sees a pair of fields that disagree only by luck.

How local outlier factor works

Neighbours	For each row, its k nearest rows (n_neighbors, 20 by default)
Density	How densely those neighbours are packed around the row
The factor	The neighbours' density over the row's own: near 1 is crowded, well above 1 is alone
Local	A row at the edge of a dense cluster can be flagged though it is near the centre overall
What comes out	A factor per row and a flag; scaled columns, because it measures distance

Duration against maturity



16

flags, k = 20

100%

recall

Precision 75%. The planted coupons and prices sit among the bonds here: LOF finds them in the other columns.

A two-column chart shows two of the five columns the screen measures.

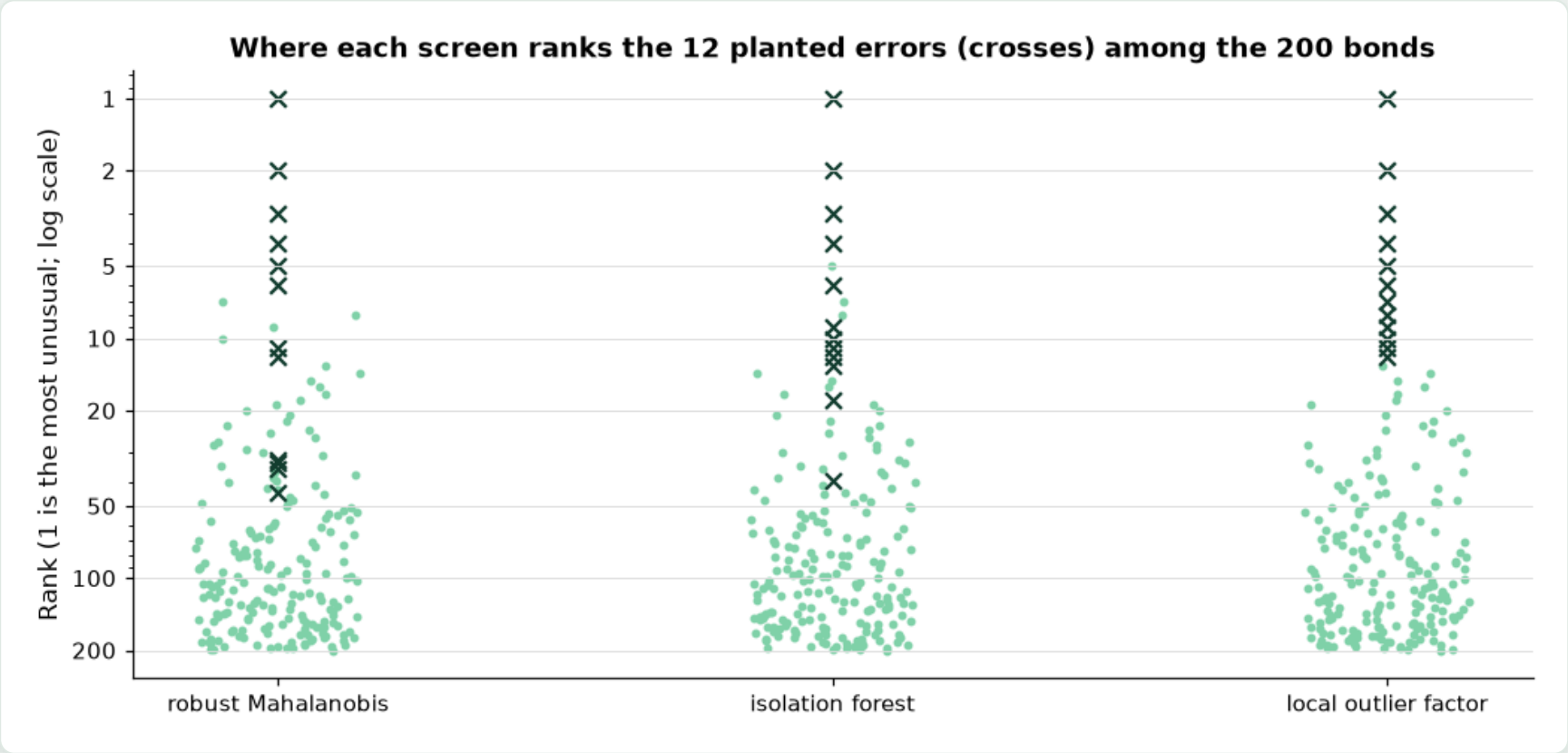
Three screens side by side

Screen	Flags	Planted found	Precision	Recall	Flags, untouched book
robust Mahalanobis, 99.9%	68	12	18%	100%	54
isolation forest, defaults	22	11	50%	92%	41
local outlier factor, k = 20	16	12	75%	100%	11

The last column is the cost of running the screen: the records it sends for checking when there is no error to find.

A chi-squared cutoff assumes a normal cloud. A bond book is not one, so the Mahalanobis screen flags many ordinary bonds.

Where each screen ranks the errors



A desk with a fixed budget checks the top k every day. Then the ranking matters more than the threshold: Exercise 3.

Contamination from the labels

12

flags, contamination 12 / 200 from the labels

75%

recall, against 92% with 'auto'

75%

precision

Set contamination to the planted share and the screen flags exactly that many: the answer, copied from the labels. A real book has no labels. Here it does not even help.

A threshold, quantile, or contamination comes from a published rule, the cost of checking, or data without labels. Never from the labels.

Few and different

Liu, Ting (Monash University), and Zhou (Nanjing University), ICDM 2008

"Isolation Forest"

Anomalies are "few and different"; a default sub-sample of 256 in their experiments

"There are only two variables in this method: the number of trees to build and the sub-sampling size." And: "A small sample size produces better iTrees because the swamping and masking effects are reduced."

Source: F. T. Liu, K. M. Ting, Z.-H. Zhou, Isolation Forest, ICDM 2008, lamda.nju.edu.cn/publication/icdm08b.pdf, the abstract and sections 1 and 4.

Excel side by side

Excel side by side (tested in Excel)

`=MINVERSE(D2:G5)`	the 4 by 4 inverse, spilled
`=MMULT(MMULT(TRANSPOSE(I2:I5-B2:B5),MINVERSE(D2:G5)),I2:I5-B2:B5)`	154.0484
`=CHISQ.INV(0.999,4)`	cutoff 18.4668

Centre in B2:B5, robust covariance in D2:G5, the day's changes in I2:I5. No isolation forest or LOF in Excel. Both values matched Python.

Save the day: find a commit by its text

```
git log --oneline -S "def mahalanobis_flags"
```

-S lists the commits whose changes added or removed that text: the one commit that brought today's function in.

On a module grown over seven days, that finds when a function arrived, and from the commit, why.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

What is drawn, from which inputs, in which unit: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the covariance where its inverse belongs: a mean of 6,528.15 in bp that moves with the units.

Code that runs is not the same as code that is right.

What you can do now

- Mahalanobis distance, classical and robust (mahalanobis_flags)
- Unusual curve days listed as history, explained by none
- Isolation forest: every setting read and swept
- LOF and its neighbourhood; three screens judged twice
- Contamination never from the labels; 30 tests pass

Week 2, Day 3 Population stability and drift